

SECURITY INCIDENT RESPONSE REPORT

Prepared by Sharvin Rao Soorithamudu

Internship Project

Table of Contents

1.0 Log Analysis	3
• 1.1 Overview of Log Files	
• 1.2 Normal Activities	
• 1.3 Failed Login Attempts	
• 1.4 Successful Logins	
• 1.5 Suspicious IP Addresses	
• 1.6 Unusual User Activities	
• 1.7 Other Abnormal Behaviour	
• 1.8 Summary of Log Analysis	
2.0 Detection Logic	7
3.0 Incident Workflow	9
4.0 Detection Logic	13
5.0 Conclusion	16

1.0 Log Analysis

1.1 Introduction

Log analysis is the process of examining system and authentication logs to identify normal operations and detect suspicious activities. Security analysts use log analysis to discover unauthorized access attempts, security incidents, and potential attacks before they cause significant damage. In this project, the provided auth.log and syslog.log files were analyzed to identify both legitimate and malicious activities.

1.2 Overview of the Log Files

Log File	Purpose
auth.log	Records authentication-related events such as successful logins, failed logins, SSH access, user sessions, and sudo activities.
syslog.log	Records operating system events, services, scheduled tasks, and application activities.

1.3 Failed Login Attempts

What was observed?

Example:

- Multiple "Failed password" entries were identified.
- Most failed attempts originated from the same IP address.
- Several different usernames were targeted.

Why is this suspicious?

Repeated failed login attempts often indicate that someone is trying to guess a user's password through a brute-force attack.

Evidence

Failed password for admin from 198.51.100.23

Failed password for backup from 198.51.100.23

Failed password for oracle from 198.51.100.23

SOC Analysis

The repeated failures from the same IP suggest an automated attack rather than a legitimate user entering an incorrect password.

1.4 Successful Logins

What was observed?

The logs contained successful authentication events.

Examples include:

Accepted publickey for aritra

Accepted password for backup

Accepted password for sysupdate

Normal vs Suspicious

Normal	Suspicious
Public key login by authorized users	Successful login immediately after numerous failed attempts
Login during expected working hours	Successful login from an attacking IP

Why is it important?

A successful login following repeated failed attempts may indicate that the attacker successfully guessed the user's password.

1.5 Suspicious IP Addresses

What was observed?

One IP address repeatedly attempted to access multiple user accounts.

Example

198.51.100.23

Why is this suspicious?

A legitimate user normally accesses only their own account.

An attacker commonly:

- Tries multiple usernames
- Performs password guessing
- Continues until a login succeeds

Possible Impact

- Unauthorized access
- Account compromise
- Data theft

1.6 Unusual User Activities

What was observed?

The logs showed attempts to access multiple accounts using the same IP.

Example usernames

- admin
- postgres
- oracle
- backup
- sysupdate

Why is this suspicious?

Normal users already know their username.

Attackers often test many common usernames before guessing passwords.

This behavior is called username enumeration.

1.7 Other Abnormal Behaviour

Examples:

- Login after many failed attempts.
- Multiple accounts accessed from one IP.
- Continuous login attempts within a short period.
- Unexpected authentication methods.
- Unusual login times (if present).

1.8 Summary of Log Analysis

Activity	Classification	Reason
CRON Jobs	Normal	Scheduled system maintenance
DHCP Renewal	Normal	Automatic network operation
Reload Nginx	Normal	Service configuration reload
Public Key Login	Normal	Secure authentication
Failed Login Attempts	Suspicious	Possible brute-force attack
Invalid Usernames	Suspicious	Username enumeration
Successful Login After Failures	Suspicious	Possible account compromise
Same IP Repeated Attempts	Suspicious	Automated attack

2.0 Detection Logic

Rule 1

Detection Rule:

If more than 5 failed login attempts occur from the same IP address within 10 minutes, generate an alert.

Reason:

Repeated failed logins often indicate a brute-force password attack.

Severity:

Medium

Rule 2

Detection Rule:

If an IP attempts to log in using more than 5 different usernames, generate an alert.

Reason:

Attackers commonly enumerate usernames before attempting password attacks.

Severity:

Medium

Rule 3

Detection Rule:

If a successful login occurs immediately after multiple failed login attempts from the same IP, generate a High severity alert.

Reason:

This may indicate the attacker successfully guessed the user's password.

Severity:

High

Rule 4

Detection Rule:

If one IP successfully logs into multiple user accounts within a short period, generate an alert.

Reason:

This behaviour may indicate compromised credentials or lateral movement.

Severity:

High

Rule 5

Detection Rule:

If an administrator account logs in outside normal working hours (for example between 12:00 AM and 5:00 AM), generate an alert.

Reason:

Administrative logins outside business hours may indicate unauthorized access.

Severity:

Medium

3.0 Incident Workflow

Incident 1

Incident Name

SSH Brute Force Attack

Description

Multiple failed login attempts from the same IP address targeting different usernames.

Severity

High

Possible Impact

- Password compromise
- Unauthorized access
- Service disruption

Recommended Response

- Block attacking IP
- Enable Fail2Ban
- Monitor SSH logs
- Notify Security Team

Incident 2

Incident Name

Username Enumeration

Description

Repeated attempts using invalid usernames.

Severity

Medium

Possible Impact

- Discovery of valid accounts
- Preparation for future attacks

Recommended Response

- Monitor activity
- Block IP if behaviour continues
- Review authentication logs

Incident 3

Incident Name

Successful Login After Failed Attempts

Description

Successful authentication occurred after repeated failed logins.

Severity

High

Possible Impact

- Account compromise
- Data theft
- Privilege escalation

Recommended Response

- Disable affected account
- Reset password
- Review account activity
- Investigate system for compromise

Incident 4

Incident Name

Multiple Accounts Accessed from Same IP

Description

One IP authenticated to multiple user accounts.

Severity

High

Possible Impact

- Credential theft
- Lateral movement
- Full system compromise

Recommended Response

- Block IP
- Isolate affected system
- Escalate to Security Team
- Review all user accounts

4.0 Detection Logic



Step 1 – Detection

Security monitoring tools identify suspicious activities from the logs, such as repeated failed logins, invalid usernames, or successful logins after many failed attempts.

Output: Suspicious activity is detected.

Step 2 – Alert

Detection rules trigger an alert in the Security Information and Event Management (SIEM) system.

The alert contains information such as:

- Source IP address
- Username
- Timestamp
- Event type
- Severity level

Output: SOC analyst is notified.

Step 3 – Investigation

The SOC analyst reviews the logs to determine whether the alert represents a genuine security incident.

The investigation includes:

- Reviewing authentication logs
- Checking login history
- Identifying affected accounts
- Correlating related events
- Assessing the scope of the incident

Output: Incident confirmed or dismissed as a false positive.

Step 4 – Response

If malicious activity is confirmed, immediate actions are taken to contain the threat.

Possible actions include:

- Blocking the attacker's IP address
- Resetting compromised passwords
- Disabling affected accounts
- Isolating compromised systems
- Escalating the incident to senior analysts

Output: Threat contained.

Step 5 – Recovery

Restore affected systems to a secure state.

Typical recovery actions include:

- Verifying system integrity
- Restoring services if needed
- Applying security patches
- Re-enabling user accounts after verification
- Confirming that no malicious activity remains

Output: Normal business operations resume securely.

Step 6 – Closure

After the incident is resolved, the SOC team documents the findings and lessons learned.

The closure process includes:

- Completing the incident report
- Recording the root cause
- Updating detection rules
- Recommending security improvements
- Closing the incident ticket

Output: Incident formally closed, with improvements in place to reduce the likelihood of similar attacks in the future.

5.0 Conclusion

This project provided practical experience in analyzing security logs to distinguish between normal and suspicious activities. Through the analysis, I identified potential security incidents such as failed login attempts, SSH brute-force attacks, suspicious IP addresses, and successful logins after multiple failed attempts. I also developed basic detection rules, classified incidents based on their severity, and recommended appropriate response actions following the incident response lifecycle.

From this project, I learned how a Security Operations Center (SOC) analyst monitors system logs, investigates security events, assesses risks, and responds to potential threats. Overall, this project strengthened my understanding of log analysis, incident detection, incident response, and the importance of documenting security incidents in a professional manner.